LOGISTIC REGRESSION

Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for classification problems**,** especially binary classification**.**

It predicts the probability that an input belongs to a particular class.

Example:
Pass/Fail, Yes/No, Spam/Not Spam, Disease/No Disease

Why It Is Called “Regression”?

Although it is used for classification, it is called regression because it uses a **regression equation** to calculate probabilities.

Working of Logistic Regression

Logistic Regression uses the **Sigmoid Function** to convert linear output into probability.

Step 1: Linear Equation
Z=W1X1+W2X2+...+bZ = W_1X_1 + W_2X_2 + ... + bZ=W1X1+W2X2+...+b

Where:

- W = Weights
- X = Inputs
- b = Bias

Step 2: Apply Sigmoid Function
σ(z)=11+e−z\sigma(z) = \frac{1}{1 + e^{-z}}σ(z)=1+e−z1

The output range is:

0 to 10 \text{ to } 10 to 1

This value represents probability.

Step 3: Decision Boundary

If:

- Probability $\geq 0.5 \rightarrow$ Class 1
- Probability $< 0.5 \rightarrow$ Class 0

Types of Logistic Regression

1.Binary Logistic Regression

- Two classes
- Example: Yes/No

2.Multinomial Logistic Regression

- More than two classes
- Example: Class A, B, C

3.Ordinal Logistic Regression

- Ordered categories
- Example: Low, Medium, High

Assumptions

- Dependent variable should be categorical
- No multicollinearity
- Independent variables can be continuous or categorical
- Large sample size preferred

Applications

- Spam detection
- Disease prediction
- Credit risk analysis
- Customer churn prediction
- Fraud detection